

IntelliScrap AI

Empowering Informal Waste Recyclers in Nigeria

INVENTOR: EDACHE MOSES ENENCHE

Overview & Mission

IntelliScrap AI is an offline-first Progressive Web App (PWA) built specifically for informal waste recyclers (*Baban Bola*). By bridging the gap between street-level collection and formal supply chains, the platform eliminates market information asymmetry using visual AI material identification, hazard warnings, and transparent pricing in a highly accessible format.

THE CHALLENGE	THE INTELLISCRAP AI SOLUTION
Information Asymmetry Middlemen exploit price opacity.	Market Transparency: Real-time, unmanipulated scrap market rates pushed via PWA, SMS, and USSD to level the playing field.
Toxic Exposure Handling hazardous e-waste blind.	Visual AI Classification: On-device ONNX models & heuristics flag toxic hazards and classify scrap instantly offline.
Digital Exclusion Low connectivity & literacy.	Multilingual Offline UX: Operates offline with Text-to-Speech (Hausa, Pidgin, English) for clear voice guidance.

Key Platform Capabilities

Smart Matching via Uber H3

Leverages hexagonal geospatial indexing to aggregate pickup points, optimizing routes and cutting logistics costs for centralized recycling hubs.

End-to-End Traceability

Tracks material flows from street collection to final processing, enabling corporate compliance with Extended Producer Responsibility (EPR) mandates.

SOCIAL & ECONOMIC IMPACT

Direct Income Growth: Democratizing pricing eliminates middleman exploitation and boosts daily earnings.

Safer Workplaces: Early toxicity warnings reduce chemical exposure and physical injuries.

Circular Integration: Converts marginalized, unconnected laborers into verified participants of the formal recycling economy.

Production-Ready PWA

DEPLOYMENT READY